The Influence of Various Seat Design Parameters: A Computational Analysis

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Abstract

Nowadays, low back pain becomes a common healthcare problem. Poor or unsuitable seat design is related to the discomfort and other healthcare problems of users. The aim of this study is to investigate the influence of seat design variables on the compressive loadings of lumbar joints. A basis that includes a musculoskeletal human body model and a chair model has been developed using LifeMOD Biomechanics Modeller. Inverse and forward dynamic simulations have been performed for various seat design parameters. The results show that the inclination of backrest and seat pan may or may not decrease the compressive spinal joint forces, depending on other conditions. The medium-level height and depth of seat pan and the medium-level and high-level height of backrest are found to cause the minimum compressive loads on lumbar joints. This work contributes to a better understanding of sitting biomechanics and provides some useful guidelines for seat design. © 2016 Wiley Periodicals, Inc.

Keywords: Musculoskeletal model; Dynamic simulation; Low back pain; Seat design; Spine biomechanics

1. INTRODUCTION

Nowadays, low back pain (LBP) is one of the most common healthcare problems of modern times and is strongly involved with the degeneration of intervertebral disc (Luoma et al., 2000). It was found that 80% of people in the United States have LBP during their lifetime (Vällfors, 1984). LBP is also the most prevalent symptom of musculoskeletal disorders in Europe (Collins & O'Sullivan, 2010). Seat design plays a very important role in the study of human sitting (Harrison, Harrison, Croft, Harrison, & Troyanovich, 1999). Poor or unsuitable seat design is related to the discomfort and other healthcare problems of users. There have been some debates about the variables of seat design in literature.

Swearingen, Wheelright, and Garner (1962) concluded that 64.8% of body weight is supported by 8% of seat area under ischia. The remaining 35.2% is for the footrests (18.4%), armrests (12.4%), and backrest (4.4%). It is believed that stability in sitting was achieved by the backrest of the seat (Swearingen et al., 1962). Many research works have been conducted to study the inclination of backrest. It was suggested that the optimal inclination should be from 90° to 125° (Kroemer, 1971; Lay & Fisher, 1940; Morant, 1947; Schede, 1935; Schulthess, 1905). The optimal seat back inclination was also researched by Knutsson, Lindh,

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Received: 24 August 2014; revised 18 September 2015; accepted 14 November 2015

View this article online at wileyonlinelibrary.com/journal/hfm

DOI: 10.1002/hfm.20649

and Telhag (1966) by the application of electromyography. The results showed that usually 110° backrest inclination is a more suitable fit to people. However, the subjects with serious disc degeneration prefer 100° backrest. It was concluded in several studies that the decrease of disc pressure is caused by the increase of backrest inclination (Andersson & Ortengren, 1974b; Andersson, Ortengren, Nachemson, & Elfstrom, 1974a, 1974b). One study about the dependency of compressive joint force with the friction coefficient in a certain backrest inclination uses musculoskeletal multibody modeling (Grujicic et al., 2010). From that study, the compressive force of L4-L5 joint increases when the coefficient decreases from 0.5 to 0.2 with a 10° backrest inclination.

The inclination of seat pan has always been a debatable topic in the past century. Initially, forward slope of seat pan was suggested by Staffel (1884). However, later in 1905, Schulthess (1905) recommended 3° to 5° backward inclination. These different seat pan inclinations were also investigated by other research groups in the following decades. It is found that there is an increase of lumbar lordosis with a forward increase of seat pan inclination (Bendix & Biering-Sørensen, 1982; Keegan, 1953). Floyd and Roberts (1958) suggested that the inclination of seat pan should be adjusted according to the job requirements. Seat pan depth that is too short will cause the thigh to hang without support and lead to fatigue (Daruis, Deros, & Nor, 2011). However, if it is too long, it will make shorter people neglect the backrest and lumbar support (Bennett, 1928; Hooton, 1970). There is also a similar situation when the seat pan is too high (Wilke, Neef, Caimi, Hoogland, & Claes, 1999). A short person who sits on a high chair will move to the edge of chair and not use the backrest or lumbar support. Therefore, dangling legs, caused by the high chair, can lead to compression stresses on the soft tissues of the posterior thigh and uncomfortable feeling to the sitting person.

In literature, there are few quantitative studies of the relationship between spinal loads and seat design parameters. Hence, the aim of this paper is to study the influence of seat design variables on the compressive loadings of spinal joints through the approach of musculoskeletal modeling and simulation using LifeMOD Biomechanics Modeller (LifeMOD). The seat design variables include backrest inclination, seat pan inclination, seat pan height, seat pan depth, and backrest height. This study can help provide some general guidelines for seat design.

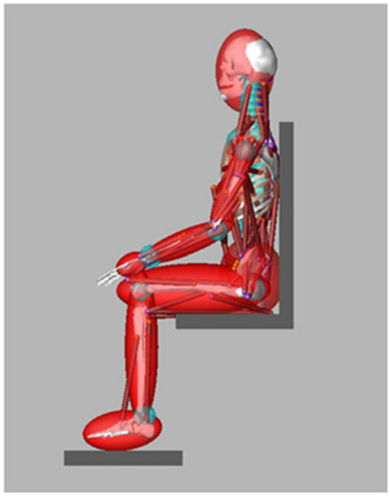

Figure 1 The basis of seated human system.

2. METHOD

A seat model was developed and integrated with the musculoskeletal multibody model in LifeMOD as shown in Figure 1 to provide a new basis for the investigation of sitting biomechanics. The musculoskeletal human body model applied in this paper has been initially proposed and validated by the author's research group (Gibson et al., 2012; Huynh, 2010; Huynh, Gibson, Jagdish, & Lu, 2015). For this musculoskeletal model, the spine is discretized into 24 individual segments of vertebra, and 25 joints are created to connect two adjacent vertebrae. The spinal joints are modeled as torsional spring forces, which are defined with user-defined properties, including stiffness, damping, angular limits, and limit stiffness values. This discretized spine model includes five types of ligaments (anterior longitudinal ligament, posterior longitudinal ligament, ligamentum flavum, capsule ligament, and interspinous ligament), four types of lumbar back muscles (multifidus muscle, erector spinae muscle, psoas major muscle, and quadrates lumborum muscle), and two types of abdominal muscles (obliquus externus and obliquus internus). In addition, the normal intra-abdominal pressure of a healthy adult, which is 20 mmHg (2.67kPa) (Cobb et al., 2005), is implemented in the musculoskeletal human body model. To validate the detailed spine model in this research, two comparison studies were carried out using LifeMOD simulation. In the first study, with the same extension moment generated in the upright position, the axial and shear forces in the L5-S1 joint calculated in the model from the inverse and forward dynamic

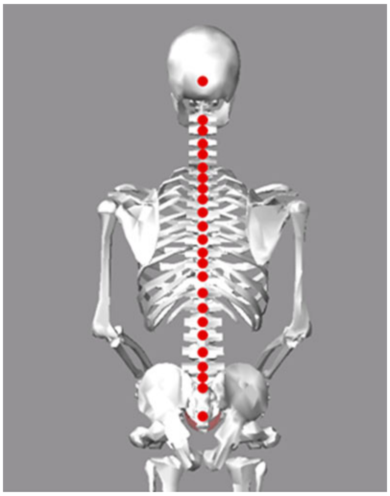

Figure 2 Contact points defined between backrest and body (back view).

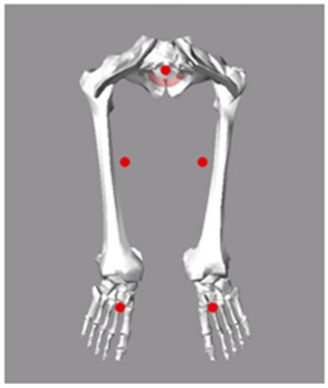

Figure 3 Contact points defined between seat pan, footrest and body (top view).

simulations were compared with those data obtained from the experiment (McGill & Norman, 1987) and another spine model (De Zee, Hansen, Wong, Rasmussen, & Simonsen, 2007). In the second study, while a human body model held a crate weighing 19.8 kg, the axial force of the L4-L5 joint was computed through the inverse and forward dynamic simulations and compared with that from the in vivo intradiscal pressure measurements (Wilke, Neef, Hinz, Seidel, & Claes, 2001). The results were consistent with findings from literature. This validated musculoskeletal model has already been applied in various applications using LifeMOD software (Gibson & Liu, 2013; Hajizadeh, Gibson, Liu, & Huang, 2012; Huang, Gibson, Lee, & Hajizadeh, 2012).

The musculoskeletal human body model used in this study was a male model (body mass: 70 kg; body height: 1.78 m) created from the GeBod anthropometric database. The body model was set to be in the upright sitting posture, with the angles of pelvis, thorax, and spine obtained through approaches of motion capture experiment (Huang et al., 2012). Meanwhile, the ratio of hip flexion and spine flexion maintains 2.2:1, as suggested by another experimental study (Bell & Stigant, 2007). Contacts were created on the sitting interface. There are 24 contact points defined between the backrest and the upper body, 3 contact points defined between the seat pan and the lower body, and 2 contact points defined between the footrest and the feet. Figures 2 and 3 show the contact points defined for the study. For better visualization, the graphics of muscles, ligaments, connecting joints, and seat are hidden in these two figures to show the locations of contact points. The contact forces are computed based on the overlapping area of the two contacting bodies and other parameters, such as stiffness, damping, coefficient of friction, and stictional and frictional transition velocities. The computational analysis is based on the inverse and forward dynamic simulations. For the inverse dynamic simulation, the musculoskeletal human body model maintains the sitting posture by recording the movements of muscles and joints. In the forward dynamic simulation, the movements of muscles and joints achieved in the inverse one were applied to drive the human-body model behaving in the same way. Because the spinal joints are created as torsional spring forces, joint forces can be obtained after the inverse and forward dynamic simulations. In this paper, the variables of seat design (Figure 4) include backrest inclination, seat pan inclination, seat pan height, seat pan depth, and backrest height. For the parameters related to the inclination of seat surface, such as backrest inclination and seat pan inclination, the effects of friction coefficient of the seat surface have also been researched.

3. RESULTS AND DISCUSSION

3.1. Backrest Inclination

Backrest inclination (Figure 4A) describes the angle between the backrest and the vertical line. The inclination angle of an upright backrest is 0°. When the backrest inclines toward the seat pan, the inclination angle is defined as negative; when the backrest inclines away from the seat pan, the inclination angle is defined as positive. Because the backrest with a positive

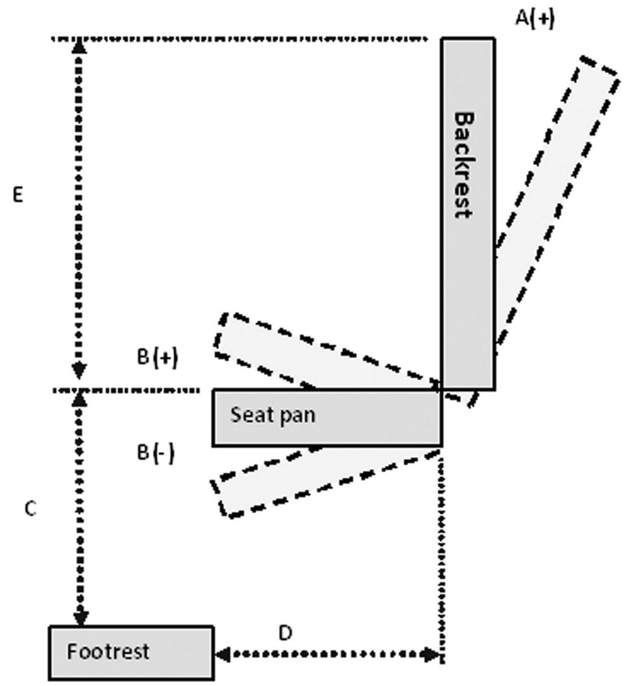

Figure 4 Variables of seat design: A = backrest inclination; B = seat pan inclination; C = seat pan height; D = seat pan depth; E = backrest height.

inclination angle is much more common in the daily life, and the friction coefficient of a common seat cover made of woven cotton fabric is around 0.4–0.6, the effects of varying backrest inclination angle from 0° to 20° in an increment of 5° and friction coefficient of seat surface from 0.2 to 0.8 in an increments of 0.2 were studied.

The variation of compressive forces of joint L2-L3, L3-L4, L4-L5, and L5-S1 over the backrest inclination and the friction coefficient are shown in Figure 5. It is found when the inclination angle increases from 0° to 20°, the compressive loads of joint L4-L5 and L5-S1 with the friction coefficients of 0.6 and 0.8 are decreased. However, greater backrest inclination can cause greater compressive forces on spinal joints in certain situations, for example, with the low friction coefficient of seat surface (0.2 or 0.4). The compressive joint force depends on the backrest inclination and the friction coefficient of seat surface in a quite complicated way.

For the discussion of backrest inclination in this part, the case with the inclination angle of 0° and the friction coefficient of 0.6 is set as the reference. Table 1 shows the relative changes of compressive joint forces in various inclination angles and friction coefficients with regard to the reference case. With the friction coefficient of 0.6, the compressive loads of joints L4-L5 and L5-S1 are reduced obviously by 9.5% on average when the inclination angle increases to 20°. This can be explained by the fact that a higher portion of body weight can be supported by the backrest as the backrest inclination increases. This finding is inconsistent with the results from other literature (Andersson & Ortengren, 1974b; Andersson et al., 1974a, 1974b).

However, a greater backrest inclination may or may not decrease the compressive joint forces, depending on other conditions, such as the friction coefficient of seat surface. When the coefficient is large enough to prevent slipping (0.6 or 0.8), the inclination angle of 20° can reduce the compressive forces on joints L4-L5 and L5-S1 by 9.5% and 12.5% on average, with the friction coefficient of 0.6 and 0.8, respectively. When the coefficient is low (0.2 or 0.4), greater inclination of backrest can cause more compressive forces on those joints. When the inclination angle changes to 20°, the compressive loads of joints are increased by 36.5% and 25%, with the friction coefficient of 0.2 and 0.4, respectively. This is due to the low friction coefficient of seat surface with low resistance to motion, resulting in a tendency of the human body to slide down from the chair. This finding is inconsistent with the results in the literature (Grujicic et al., 2010) and supports the conclusion that the lumbar disc pressure is higher if the lumbar lordosis is maintained by muscle activity other than back support (Andersson & Ortengren, 1974a).

3.2. Seat Pan Inclination

Seat pan inclination (Figure 4B) describes the angle between the seat pan and the horizontal line. The inclination angle of a horizontal seat pan is 0°. When the seat pan inclines toward the floor, the angle is defined as positive; when the seat pan inclines away from the floor, the angle is defined as negative. In this study, the effects of varying seat pan inclination angle from −10° to 10° in an increment of 5° and varying friction coefficient of seat surface from 0.2 to 0.8 in an increment of 0.2 are examined.

The variation of the compressive force of joints L2-L3, L3-L4, L4-L5, and L5-S1 over different seat pan inclinations and friction coefficients are shown in Figure 6, with the positive inclination defined as the forward tilt of seat pan. In this study, all these joints have maximum compressive forces at the seat pan inclination angle of −10° and minimum at 10°. The effects

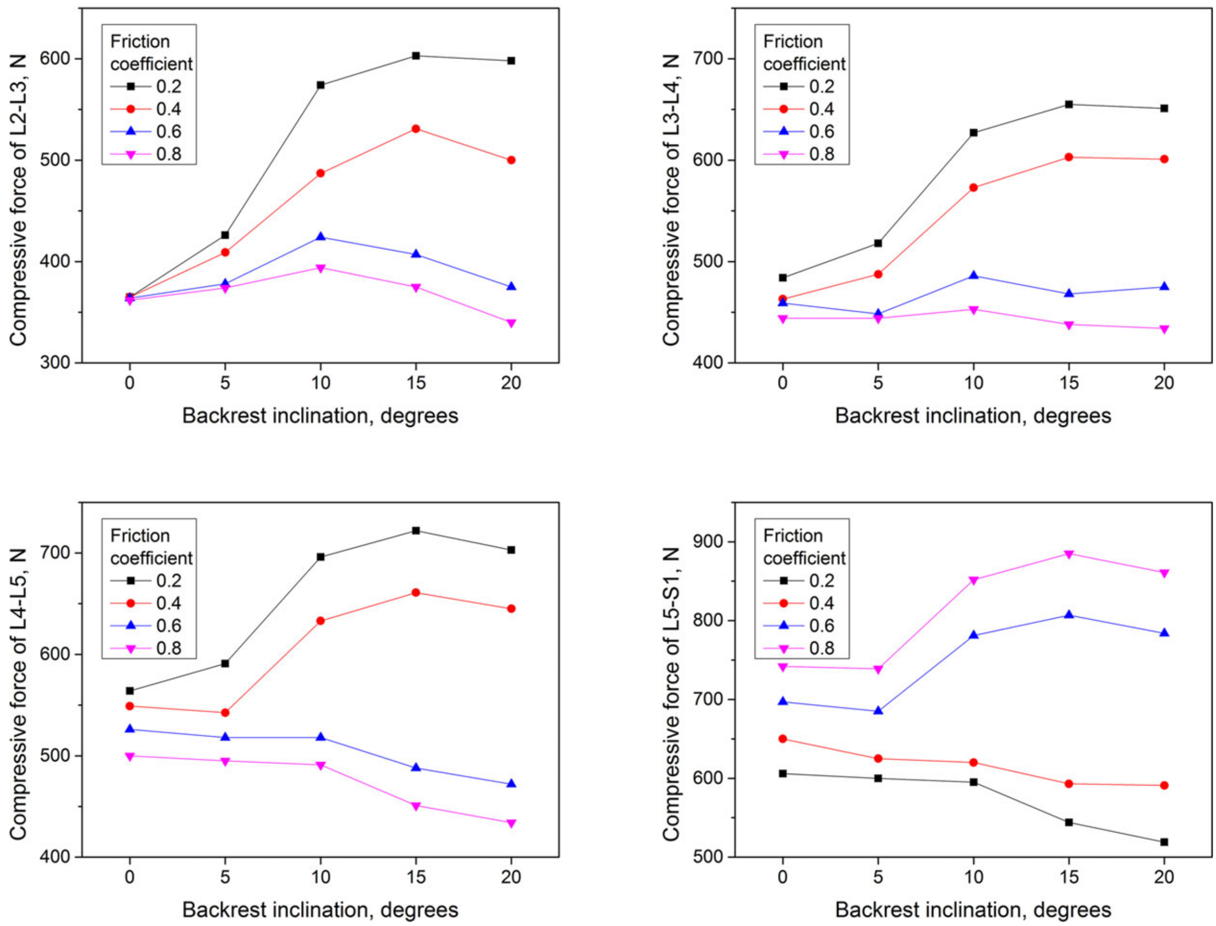

Figure 5 The variation of compressive forces of lumbar joints over the backrest inclination.

of friction coefficient on spinal loads are not significant with the forward inclination of seat pan but quite significant with the backward inclination of seat pan. It is obvious that, with the backward seat pan inclination, the compressive forces of spinal joint with friction coefficient 0.2 are higher than those with friction coefficient 0.4.

For the discussion of seat pan inclination in this part, the case with the inclination angle of 0° and the friction coefficient of 0.6 is set as the reference case. Table 2 shows the relative changes of compressive joint forces in various inclination angles and friction coefficients with regard to the reference case. As discussed before, there are some disagreements on the optimal inclination of seat pan in literature. From the results of the current study, it is found that the forward tilt of seat pan can decrease the compressive forces of joints by an average of 20.6% and 30.6% with the inclination angle of 5° and 10°, respectively, which is beneficial to the human body. It was expected that the backward inclination of seat pan would increase the load at the lumbar spine due to the reduction of lordosis or the appearance of kyphosis of the lumbar spine when the thigh-trunk angle decreases from 200° to 50° (Keegan, 1953). However, it is found that the backward tilting seat pan may or may not decrease the compressive force of spinal joints, depending on the inclination angle. A backward seat pan inclination of −10° can increase the compressive loading of lumbar spinal joints by an average of 12.9%, but a backward inclination of −5° can reduce the compressive forces of joints by an average of 17.1% compared with no inclination. The support from the seat backrest may be a reason for the decrease of compressive forces with the backward seat pan inclination of −5°. When the seat pan tilts backward, the human body is also pushed backward, which forms better contact between the back and the backrest.

3.3. Seat Pan Height

Seat pan height (Figure 4C) is the distance from the seat pan to the footrest. The medium-level seat pan height,

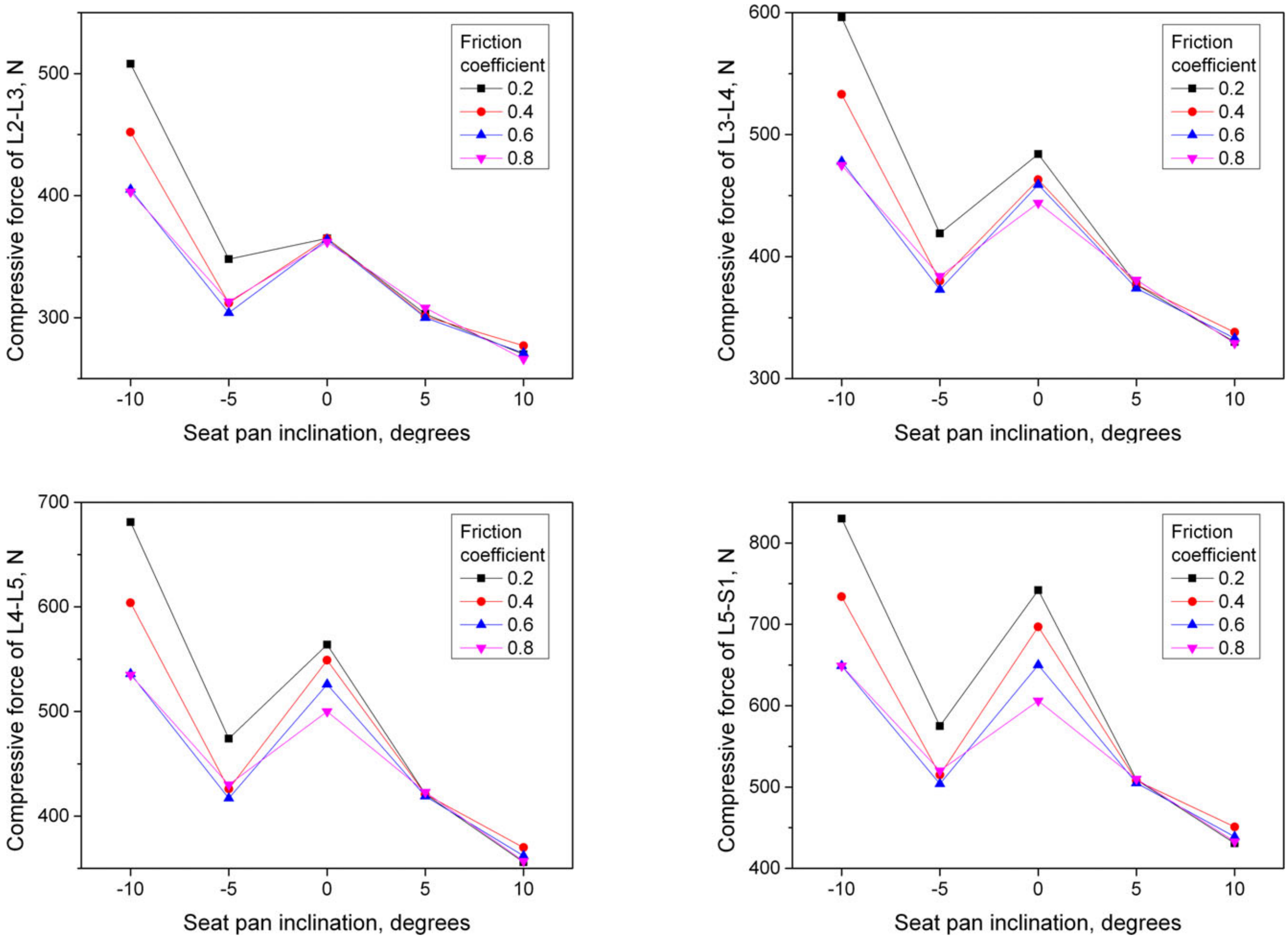

Figure 6 The variation of compressive forces of lumbar joints over the seat pan inclination.

which should be adjusted according to the height of the sitting subject, is defined to enable a 90° knee angle. In this study, the medium-level seat pan height is 463 mm for the sitting body model; the low-level seat pan height is set as 363 mm, with less than 90° knee angle; while the high-level seat pan height is set as 563 mm, with a 90° knee angle and dangling feet of the sitting body model. In this study, the effects of seat pan with low-level height, medium-level height, and high-level height are examined.

The effects of varying seat pan heights on the compressive forces of lumbar joints from L2-L3 to L5-S1 are shown in Figure 7. From the figure, it is found that the medium-level seat pan height leads to the minimum compressive loads on the lumbar joints. The compressive forces with different seat pan heights increase in the following order: medium level, high level, and low level.

When the human body sits on the seat pan with the medium-level height, which means the knee angle is 90° and the feet are in contact with the footrest, the seat is able to provide good support to the lower torso, upper legs, and feet. Table 3 shows the relative changes of compressive joint forces in low-level and high-level seat pan height with regard to the medium-level seat pan height (i.e., the reference case). Compared with the medium-level height, the average increase of compressive spinal load with the low-level height is around 25.5%. This is because, with the low-level height of seat pan, only parts of upper legs are supported by the seat pan. Thus, a higher than usual portion of the body weight is transferred to the lower torso and the feet, since the knee angle of the sitting body is less than 90°. As the knee angle decreases, the lumbar curvature changes from lordosis to kyphosis (Keegan, 1953), which can increase the compressive forces on the lumbar joints (Harrison et al., 1999).

However, the average increase of compressive spinal load with the high-level height is around 14.0% compared with the medium-level height. Although the

TABLE 1. Relative Changes of Compressive Joint Forces in Various Backrest Inclination Angles and Friction Coefficients with Regard to the Reference Case

		Joint L2-L3				Joint L3-L4				Joint L4-L5				Joint L5-S1			
		friction coefficient				friction coefficient				friction coefficient				friction coefficient			
		0.2	0.4	0.6	0.8	0.2	0.4	0.6	0.8	0.2	0.4	0.6	0.8	0.2	0.4	0.6	0.8
Inclination angle	−10°	+40%	+24%	+11%	+11%	+30%	+16%	+4%	+3%	+29%	+15%	+2%	+2%	+28%	+13%	0%	0%
	−5°	−4%	−14%	−16%	−14%	−9%	−17%	−19%	−16%	−10%	−19%	−21%	−18%	−12%	−21%	−22%	−20%
	0°	0%	0%	Ref.	−1%	+5%	+1%	Ref.	−3%	+7%	+4%	Ref.	−5%	+14%	+7%	Ref.	−7%
	5°	−17%	−17%	−18%	−15%	−18%	−18%	−19%	−17%	−20%	−20%	−20%	−20%	−22%	−22%	−22%	−22%
	10°	−26%	−24%	−26%	−27%	−28%	−26%	−27%	−28%	−32%	−30%	−31%	−32%	−34%	−31%	−32%	−33%

TABLE 2. Relative Changes of Compressive Joint Forces in Various Seat Pan Inclination Angles and Friction Coefficients with Regard to the Reference Case

		Joint L2-L3				Joint L3-L4				Joint L4-L5				Joint L5-S1			
		friction coefficient				friction coefficient				friction coefficient				friction coefficient			
		0.2	0.4	0.6	0.8	0.2	0.4	0.6	0.8	0.2	0.4	0.6	0.8	0.2	0.4	0.6	0.8
Inclination angle	0°	0%	0%	Ref.	−1%	+5%	+1%	Ref.	−3%	+7%	+4%	Ref.	−5%	+14%	+7%	Ref.	−7%
	5°	+17%	+12%	+4%	+3%	+13%	+6%	−2%	−3%	+12%	+3%	−2%	−6%	+14%	+5%	−4%	−8%
	10°	+58%	+34%	+16%	+8%	+37%	+25%	+6%	−1%	+32%	+20%	+1%	−7%	+31%	+20%	−5%	−8%
	15°	+66%	+46%	+12%	+3%	+43%	+31%	+2%	−5%	+37%	+26%	−7%	−14%	+36%	+24%	−9%	−16%
	20°	+64%	+37%	+3%	−7%	+42%	+31%	+3%	−5%	+34%	+23%	−10%	−17%	+32%	+21%	−9%	−20%

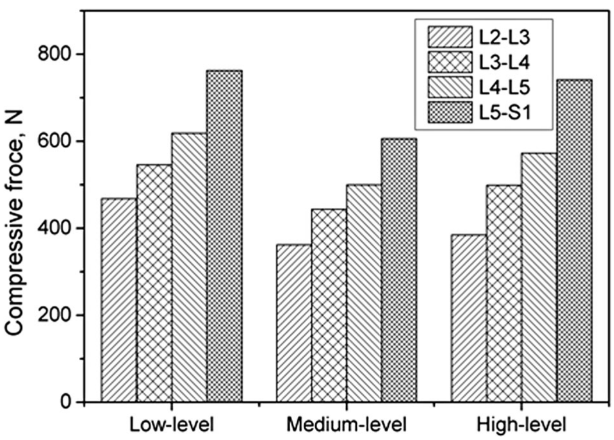

Figure 7 The variation of compressive forces of lumbar joints over the seat pan height.

TABLE 3. Relative Changes of Compressive Joint Forces in Various Seat Pan Heights with Regard to the Reference Case

	Joint L2-L3	Joint L3-L4	Joint L4-L5	Joint L5-S1
Low level	+29.3%	+23.0%	+23.8%	+25.9%
High level	+6.4%	+12.4%	+14.6%	+22.4%

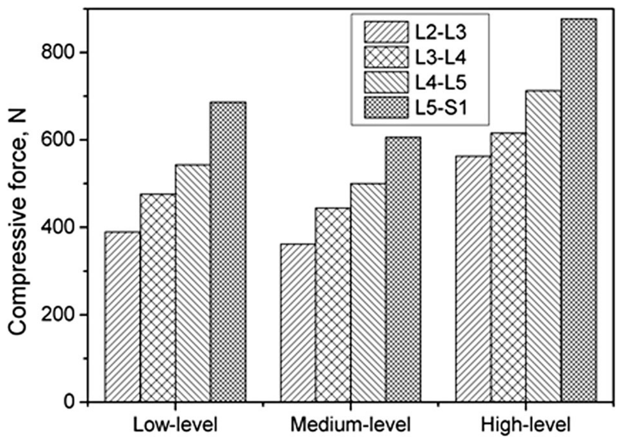

Figure 8 The variation of compressive forces of lumbar joints over the seat pan depth.

TABLE 4. Relative Changes of Compressive Joint Forces in Various Seat Pan Depths with Regard to the Reference Case

	Joint L2-L3	Joint L3-L4	Joint L4-L5	Joint L5-S1
Low level	+7.7%	+7.2%	+8.6%	+13.2%
High level	+55.5%	+38.7%	+42.6%	+44.7%

seat pan with the high-level height can maintain the same knee angle (90°) as the medium-level height, it can incur the dangling feet of the sitting body (Keegan, 1953). This leads to the neglect of support from footrest, which is able to bear 18.4% of the body weight (Swearingen et al., 1962). Thus, the seat pan with the high-level height causes greater compressive loads on the lumbar joints as compared with the medium-level height seat pan.

3.4. Seat Pan Depth

Seat pan depth (Figure 4D) is the distance from the front edge to the back edge of the seat pan in the sagittal plane of sitting body. The seat pan with the medium-level depth can support almost full upper legs with four fingers space (70 mm) between the knees and the front edge of the seat pan and allow the sitting body to make use of the backrest. In this study, the medium-level depth of the seat pan is set as 400 mm. The low-level depth is set as 300 mm, with a distance of 170 mm between the knees and the front edge of the seat pan. The high-level depth is set as 500 mm, with a distance of 70 mm between the knees and the front edge of the seat pan; thus, the sitting body is not able to use the backrest to support the upper torso. In this study, the effects of seat pan with low-level depth, medium-level depth, and high-level depth are examined.

The influence of different seat pan depths on the compressive forces of lumbar joints from L2-L3 to L5-S1 is shown in Figure 8. From the figure, the compressive forces with different seat pan depths increase in the following order: medium level, low level, and high level. The seat pan with the high-level depth leads to highest compressive loads on the lumbar joints than the other two levels.

The relative changes of compressive joint forces in low-level and high-level seat pan depth with regard to the medium-level seat pan depth (i.e., the reference case) are shown in Table 4. Compared with the medium-level depth, the seat pan with the low-level depth is not able to provide full support for the upper legs. In this case, more energy is required to maintain the sitting posture; thus, the compressive load in the low-level depth is around 9.2% higher than the medium-level depth. When the depth of seat pan is at the high level, the body can only sit on the frontal part of the seat pan, with full support for upper legs. However, this type of seat pan depth causes the sitting body to neglect the backrest. Since there is no contact

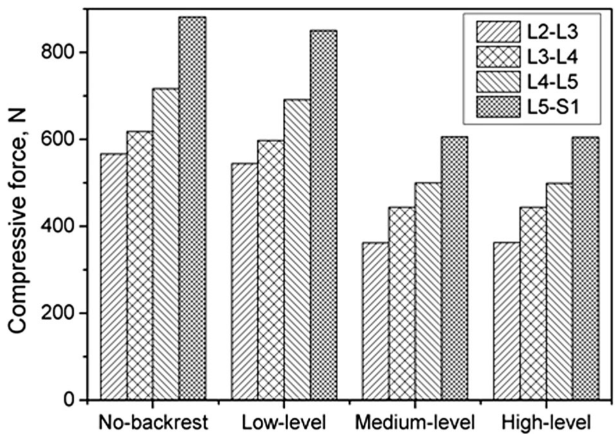

Figure 9 The variation of compressive forces of lumbar joints over the backrest height.

between the body and the backrest, the human body needs to maintain the lumbar curvature by its own muscle activation instead of the support from backrest. This explains why the high-level seat pan depth causes significantly higher compressive forces by 45.4% on average than the medium-level seat pan depth. This result is also in agreement with other studies (Andersson & Ortengren, 1974a; Wilke et al., 2001) with the conclusion that the lumbar disc pressure with lumbar lordosis maintained by the muscle activity is higher than the one maintained by the lumbar support.

3.5. Backrest Height

Backrest height (Figure 4E) is the distance from the top edge to the bottom edge of the backrest in the sagittal plane of sitting body. The backrest with the medium-level height means it is able to provide the full shoulder support for the sitting body, which is 650 mm in this study. The low-level height is defined as the ability to support only the lumbar region, which is set as 300 mm. The high-level height is defined as the ability to fully support the head and neck, which is set as 950 mm. Seats without backrest are another common type of seats in daily life. Thus, in this study, the effects of seat without backrest, and with low-level, medium-level, and high-level backrest heights are examined.

Figure 9 shows the results of compressive forces of lumbar joints over the different backrest heights. Four types of backrests were included: 1) no-backrest, 2) low-level height, 3) medium-level, and 4) high-level backrest heights. From the figure, the seats with no backrest and the low-level backrest height cause higher compressive forces on lumbar joints than the medium-level and the high-level backrest heights.

TABLE 5. Relative Changes of Compressive Joint Forces in Various Backrest Heights with Regard to the Reference Case

	Joint L2-L3	Joint L3-L4	Joint L4-L5	Joint L5-S1
No backrest	+4.0%	+3.3%	+3.6%	+3.6%
Medium level	−33.6%	−25.8%	−27.7%	−28.8%
High level	−33.4%	−25.8%	−27.9%	−28.9%

Table 5 shows the relative changes of compressive joint forces with no-backrest, medium-level, and high-level backrest heights with regard to the low-level backrest height (i.e., the reference case). In this study, the seat with no-backrest causes the highest compressive forces on the lumbar joints, with an average of 3.6% higher than the low-level backrest height. The reason is the sitting body has to maintain the spine curvature by its muscle activity, thus leading to higher compressive loads on the lumbar joints (Andersson & Ortengren, 1974a). For the backrest with the low-level height, it can only provide support for the lumbar part of spine. Hence, the compressive loads of lumbar joints are lower than the seat without backrest but higher than the seat with the medium-level height. The backrest with the medium-level height can provide support from the shoulder to the lower torso and help relax the muscles on the back and reduce the energy expenditure. Hence, the compressive load of lumbar joints with the medium-level backrest height is around 29.0% lower than the low-level backrest height. It was expected that the backrest with the high-level height can lead to the minimum compressive loads, since it supports the body from head to lower torso. However, the compressive forces with the high-level height and the medium-level height are quite similar. This may be explained by the fact that the placement of the backrest in this study is upright with 0° inclination angle. Because of the natural curvature of the spine, when people sit upright on the seat with the high-level backrest height, little or even no contact exists between the head and the backrest.

3.6. Validation

In literature, some research works have been conducted to investigate the relations between spinal loads and postures through in vivo pressure needle measurements (Nachemson, 1966, 1981; Nachemson & Morris, 1964; Sato, Kikuchi, & Yonezawa, 1999; Wilke et al., 1999; Wilke et al., 2001). The experiment of direct measurement of disc pressure is most suitable for the validation of the results of the current study. However, because of the invasive effect of the direct measurement, it is difficult to conduct such experiment for direct validation at the moment.

Hence, in this paper, a comparison is made with in vivo intradiscal pressure measurements of the L4-L5 disc as reported by Wilke et al. (2001) in which the experiment participant (body mass: 70 kg; body height: 1.74 m) shows similar body height and weight to the musculoskeletal body model in the current study. A pressure of 0.44 MPa in the L4-L5 disc was measured while the participant was in the posture of relaxed erect sitting without backrest. The disc cross-sectional area of the participant was given as 18 cm^2 in literature and the axial force can be calculated as 792 N. The same situation of sitting with no backrest has been investigated using the musculoskeletal human body model as previously described in the part of backrest height in this paper. The estimated axial force in upright sitting with no backrest was 727 N. This is a quite close match, considering that no attempt was made to scale the model to the experiment participant in this study.

4. CONCLUSION

In summary, this paper provides a quantitative study on the influence of variable seat design parameters, including backrest inclination, seat pan inclination, seat pan height, seat pan depth, and backrest height, on the compressive forces on lumbar joints through the musculoskeletal modeling and simulation. It is found that the sitting biomechanics is very complicated: The inclination of backrest and seat pan may or may not decrease the compressive joint forces, depending on other conditions. The medium-level height and depth of seat pan and both the medium-level and high-level backrest heights are found to cause the minimum compressive loads on lumbar joints. Generally, this work contributes to a better understanding of sitting biomechanics and provides some general rules of seat design and some guidelines for the optimal seat design parameters.

Acknowledgments

This research was supported by the Fundamental Research Funds for the Central Universities and the Singapore Ministry of Education Academic Research Fund (AcRF).

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